Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Glucose molecule reacts with X number of molecules of phenylhydrazine to yield osazone. The value of X is

(A) three

(B) two

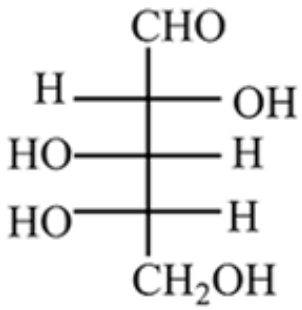
(C) one

(D) four

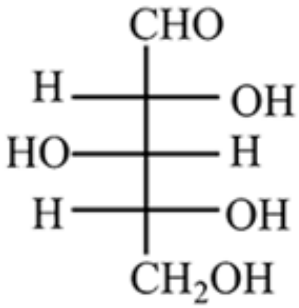
Q.2

Which L-sugar on oxidation gives an optically active dibasic acid (2 COOH groups)?

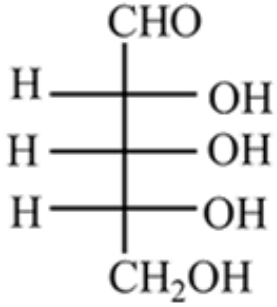
(A)



(B)

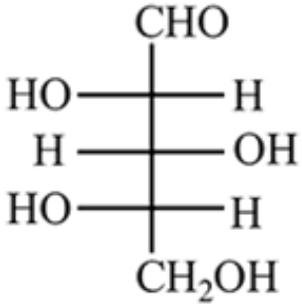


(C)



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(D)



Q.3

In cells the net production of ATP molecules generated from one glucose molecule is

(A) 46

(B) 32

(C) 36

(D) 40

Q.4

Complete hydrolysis of starch gives :

(A) glucose only

(B) galactose and fructose in equimolar amounts

(C) glucose and galactose in equimolar amounts

(D) glucose and fructose in equimolar amounts

Q.5

In osazone formation, glucose reacts with three molecules of phenylhydrazine. Which statement is true regarding this?

(A) All the three molecules react in similar fashion

(B) Two molecules react in similar manner, while the third reacts in different way

(C) All the three molecules react in different ways

(D) None of the above is true

Q.6

What will happen when $\mathrm{D}-(+)$ -glucose is treated with methanolic $-\mathrm{HCl}$ followed by Tollens' reagent?

(A) A black ppt. will be formed

(B) A red ppt. will be formed

(C) A green colour will appear

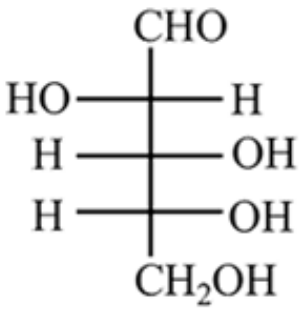
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(D) No characteristic colour or ppt. will be formed.

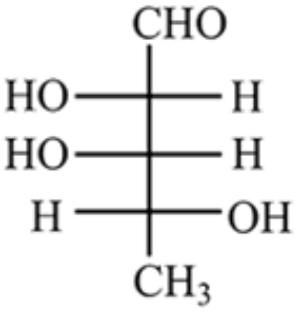
Q.7

Among the three compounds shown below, two yield the same product on reaction with warm HNO_3. The exception is:

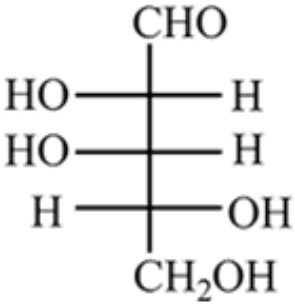
(A)



(B)



(C)



(D) None of these

Q.8

$D-(+)$ -Glucose $\xrightarrow{5(CH_3CO)_2O}$ $D-(+)$ -Glucose pentaacetate

Which statement is true about glucose pentaacetate?

(A) It will react with phenylhydrazine but not with Tollens' reagent.

(B) It will react with Tollens' reagent but not with phenylhydrazine.

(C) It will react with both of the above mentioned reagents.

(D) It will react neither with phenylhydrazine nor with Tollens' reagent.

Q.9

The two forms of D-glucopyranose obtained from the solution of D-glucose are called

(A) Isomer

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(B) Anomer

(C) Epimer

(D) Enantiomer

Q.10

The number of molecules of ATP produced per molecule of glucose in glycolysis is

(A) 38

(B) 36

(C) 18

(D) 2

Q.11

Which is correct statement?

(A) Starch is a polymer of α -glucose

(B) In cyclic structure of fructose, there are four carbons and one oxygen atom

(C) Amylose is a component of cellulose

(D) Proteins are composed of only one type of amino acids

Q.12

Which of the following statement is not correct?

(A) Amylopectin is a branched polymer of α glucose.

(B) Cellulose is a linear polymer of β -glucose

(C) Glycogen is the food reserve of plants

(D) All proteins are polymers of α -amino acids.

Q.13

Which of the following is a non-reducing sugar?

(A) Lactose

(B) Fructose

(C) Sucrose

(D) Maltose

Q.14

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Which of the following does not reduce Benedict's solution?

(A) Glucose

(B) Fructose

(C) Sucrose

(D) Aldehyde

Q.15

Hydrolysis of sucrose is called

(A) hydration

(B) saponification

(C) esterification

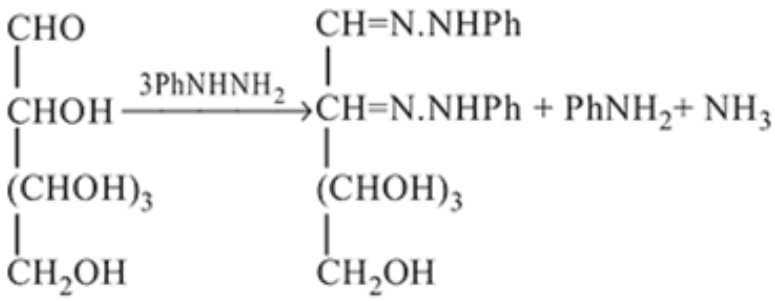
(D) inversion

Answers & Solutions

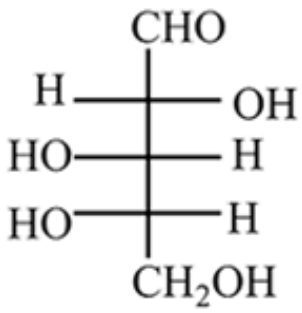
Q.1 Answer:

three

Solution:



Q.2 Answer:



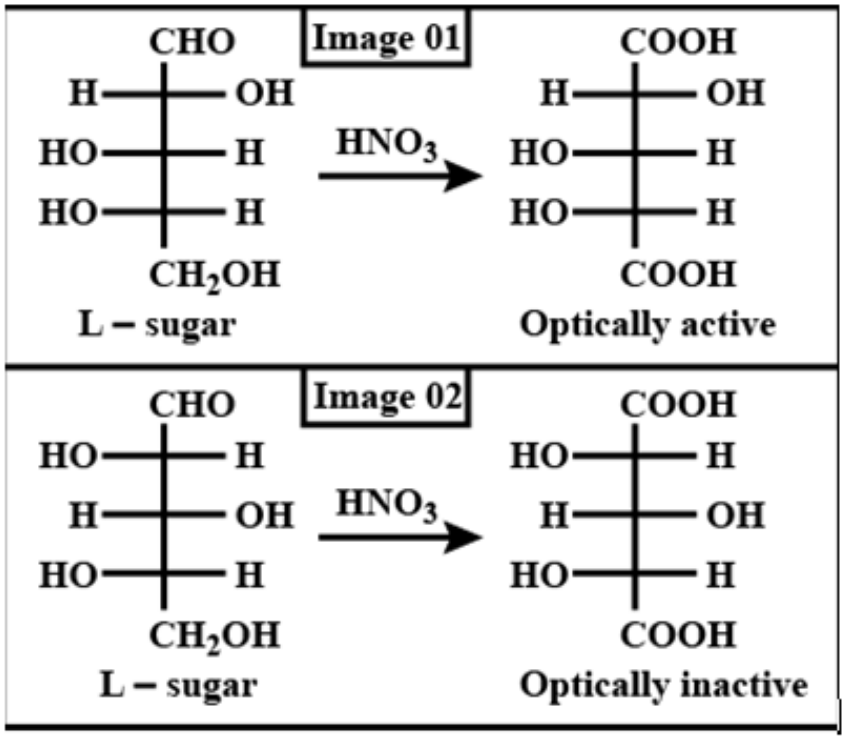
Solution:

Any sugar will be optically active when it does not have any plane of symmetry. As only (a) \& (d) are L-sugar, (b) \& (c) are D-sugar after oxidation.

(a) (Refer to Image 1) - The product do not superimpose in its mirror image.

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(d) (Refer to Image 2) - As it has plane of symmetry.



Q.3 Answer:

36

Solution:

$C_6H_{12}O_6 + 6O_2 + 2ATP \longrightarrow$
$6CO_2 + 6H_2O + 38ATP$
$\therefore$ Net ATP molecules evolved $= 36$

Q.4 Answer:

glucose only

Solution:

Starch is a polymer of $\alpha - D$**-glucopyranose.**

Complete hydrolysis of starch gives glucose only.

$$\underset{\text{starch}}{(C_6H_{10}O_5)_n} + nH_2O \xrightarrow[393K, 2-3atm]{dilH_2SO_4} \underset{\text{glucose}}{nC_6H_{12}O_6}$$

Q.5 Answer:

Two molecules react in similar manner, while the third reacts in different way

Solution:

First molecule of phenylhydrazine unedrgoes nucleophilic addition on carbonyl (- CHO in glucose and $> CO$ **in fructose) group. Second molecule of the reagent oxidizes** $-CHOH$ **- at position 2 (in aldoses) or** $-CH_2OH$ **at position 1 (in ketoses) to form** $-CHO$ **or** $> C = O$ **respectively. The**

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third molecule again undergoes nucleophilic addition on the newly developed carbonyl group to form osazone.

Q.6 Answer:

No characteristic colour or ppt. will be formed.

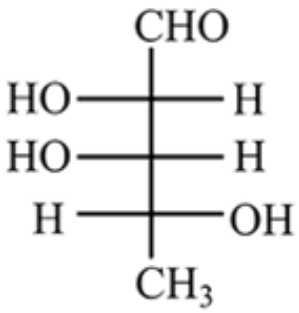
Solution:

When $D-(+)$ - glucose is treated with methanolic $-HCl$ followed by Tollens' reagent, no characteristic colour or ppt. will be formed.

Reaction of $D-(+)$ glucose with methanolic $-HCl$ leads to formation of methyl glucoside (-OH group at C_1 of hemiacetal is methylated to form acetal). Acetal, is not hydrolysable by base, so it will not respond Tollen's reagent.

Note: Tollen's reagent oxidises aldehyde to carboxylate group. Tollen's reagent itself is reduced to black silver mirror. If aldehdye group is not free (such as converted to acetal), this reaction will not occur.

Q.7 Answer:



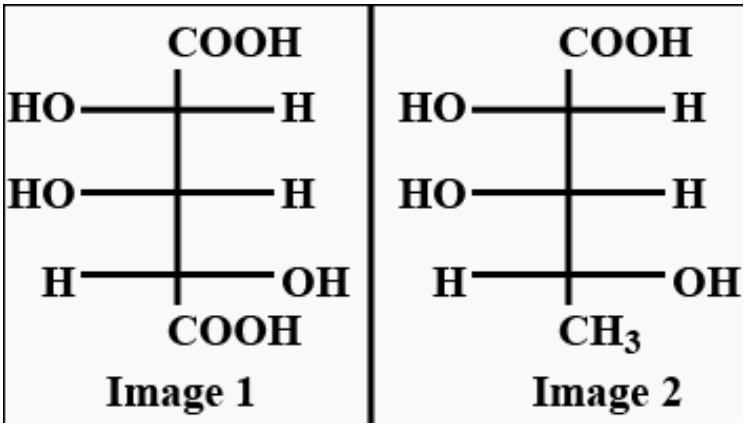
Solution:

On reaction with HNO_3 oxidation of aldehyde and alcohol takes place which gives carboxylic acid. But methyl do not oxidise with HNO_3. Thus (a) $\&(c)$ will give same product i.e.

(Refer to Image 1)

But (b) on oxidation give

(Refer to Image 2) which is different product than that given by $(a)\&(c)$.



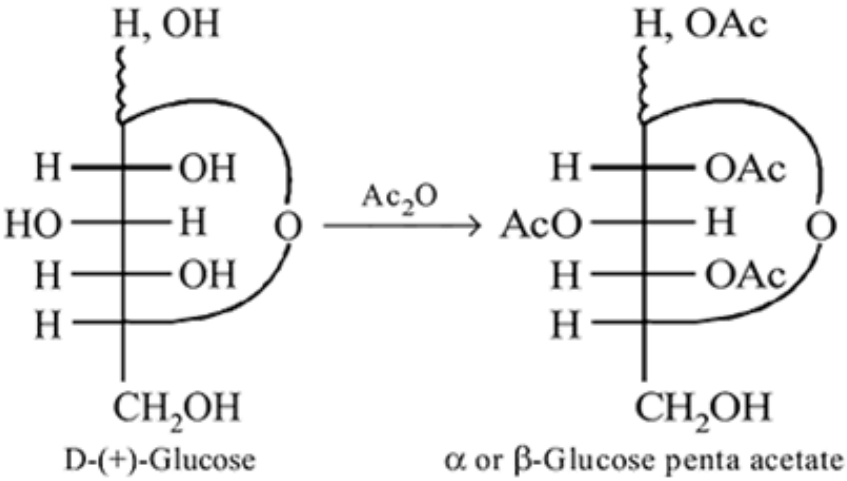
Q.8 Answer:

It will react neither with phenylhydrazine nor with Tollens' reagent.

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Solution:

During acetylation of $(+)$ -glucose, it is the $C_1 - OH$ of the hemiacetal that is acetylated and not the $C_5 - OH$ that forms the ring (cyclic structure). Since equilibrium with the open-chain aldehyde is prevented, the penta-acetate does not respond the aldehydic reactions.



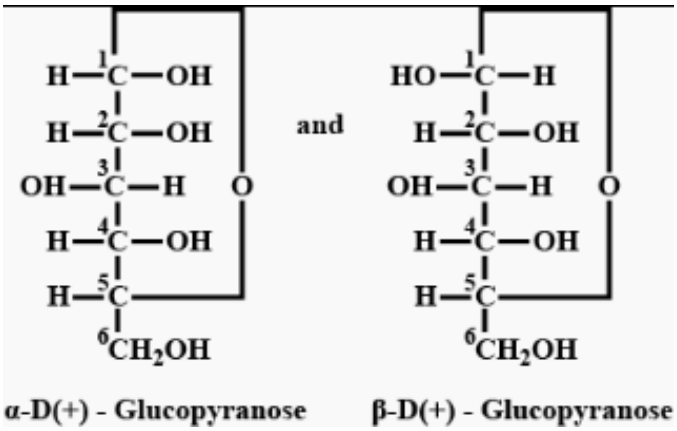
D-(+)-Glucose α or β-Glucose penta acetate

Q.9 Answer:

Anomer

Solution:

Two forms of D-glucopyranose are $\alpha - D - (+)$-glucopyranose and $\beta - D -$ (+)-glucopyranose. These are anomers (a pair of stereoisomers) which differ in configuration only around first carbon atom are called anomers.



α-D(+) - Glucopyranose β-D(+) - Glucopyranose

Q.10 Answer:

2

Solution:

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Product	Molecule	ATP
Glycolysis	2 ATP 2NADH	8ATP
Krebs cycle NADH	3 (in two cycle)	18 ATP
$FADH_2$	4 (in two cycle)	4 ATP
GTP	2 (in two cycle)	2ATP
Oxidative decarboxylation	2 NADH (in two cycle)	6ATP
Total		38

The number of molecules of ATP produced per molecule of glucose in glycolysis is 2 .

Q.11 Answer:

Starch is a polymer of α -glucose

Solution:

Starch is also known as amylum which occurs in all green plants. A molecule of starch $(C_6H_{10}O_5)_n$ is built of a large number of α -glucose rings joined through oxygen atom.

Q.12 Answer:

Glycogen is the food reserve of plants

Solution:

Glycogen is called animal starch and is found in all animal cells. It constitutes the reserve food material

Q.13 Answer:

Sucrose

Solution:

Sucrose is a non-reducing sugar as it does not reduce Fehling reagent and Tollen's reagent.

Q.14 Answer:

Sucrose

Solution:

Sucrose, being a non-reducing sugar, does not reduce Benedict's solution. Remember that fructose has an α -hydroxy ketonic group, which is also reducing group (different from ordinary ketonic group)

Q.15 Answer:

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inversion

Solution:

Since sucrose is dextrorotatory while hydrolysis product of sucrose, having equimolar mixture of glucose and fructose, is laevorotatory. Hence the hydrolysed product of sucrose is known as invert sugar and the hydrolysis of sucrose is known as inversion.

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